

Two-Level Silicon Photonic Wavelength Demultiplexer: Cascaded MZI Lattice Filters and Ring Resonators, Benchmarked Against an AWG

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Fabrication: SiEPIC openEBL, 220 nm SOI, cell $605 \times 410 \mu\text{m}$

Abstract

Fibre capacity growth increasingly relies on dense wavelength-division multiplexing (WDM). However, no single integrated silicon photonics filter provides both a wide, flat passband for coarse band selection and a sub-nanometer passband for fine channel selection. To solve this, we propose a two-level wavelength demultiplexer: a 1×4 MZI lattice filter tree for coarse 10 nm channel splitting (1535–1565 nm), cascaded with microring resonators (FSR 12.1 nm, FWHM 0.2 nm) for fine 200 GHz channel selection. Contra-directional couplers (CDCs) are integrated at the ring drops to filter out spurious ring orders, removing the free-spectral-range limit. The design is rigorously validated through Lumerical simulations and benchmarked against standard arrayed waveguide gratings (AWGs).

Keywords: Silicon photonics, wavelength-division multiplexing, MZI lattice filter, microring resonator, contra-directional coupler, arrayed waveguide grating

Notation

Symbol	Meaning
λ, λ_0	free-space wavelength; expansion point (1550 nm)
n_{eff}, n_g	effective index, group index of a guided mode
$\beta = 2\pi n_{\text{eff}}/\lambda$	propagation constant
FSR	free spectral range (wavelength period of a periodic filter)
ΔL	differential or unit path length
$K = \sin^2 \theta$	power cross-coupling of a coupler; θ = coupling angle
r, κ	field self- and cross-coupling of a ring coupler, $r^2 + \kappa^2 = 1$
a	ring single-pass field transmission
Λ, N	grating period, number of periods
κ_g	grating (contra-directional) coupling coefficient
m	integer order (interference or diffraction)
n_s, R_a, d, d_o, M	AWG: slab index, focal length, arm pitch, output pitch, number of arms
C1–C4, F1–F4	coarse and fine channels

1 Introduction

1.1 The State of the Art in High-Speed Data Transfer

The Core Idea: If we want to push more data through a single optical fiber, we essentially have five parallel dimensions to play with. But here’s the catch—wavelength multiplexing is the foundational dimension that multiplies all the others. Let’s break down how the industry is currently tackling this.

1. Coherent Modulation (Dual-Polarization)

- **The Mechanism:** Instead of just turning a laser on and off, coherent transceivers encode data into both the amplitude and the phase of the light (using QAM—Quadrature Amplitude Modulation), and they do this on two orthogonal polarizations. A receiver then uses a local oscillator and digital signal processing to untangle the signal.
- **Where the Industry is At:** As of late 2024, the OIF 800ZR standard locked in dual-polarization 16-QAM running at 118.2 GBd. This gives us 150 GHz channel spacing, ideal for connecting data centers over 80–120 km [1].
- **The Bottleneck:** We can’t just keep cranking up the symbol rate or the constellation size. We eventually hit the ceiling of electronic bandwidth and signal-to-noise ratios. To scale further, we don’t necessarily need *faster* channels; we need *more* of them.

2. Wavelength-Division Multiplexing (WDM)

- **The Mechanism:** Think of WDM as a multi-lane highway. It packs multiple independent light carriers into a single fiber, each on its own wavelength. To make this work, you need highly precise filters at both ends to combine (mux) and separate (demux) the channels.
- **The Grids:** Long-haul line systems pack them tight using the ITU-T dense grid (DWDM, 50 or 100 GHz spacing) [2], while data centers typically use the looser coarse grid (CWDM, 20 nm spacing) [2].
- **Why It’s Central to This Design:** WDM is the exact dimension this proposal targets. Developing the on-chip multiplexer/demultiplexer filter *is* the hardware challenge we’re solving.

3. Expanding the Spectrum (Multi-Band Transmission)

- **The Mechanism:** When the standard C-band gets too crowded, we spill over into the L, S, E, and O bands to tap into the fiber’s remaining low-loss windows.
- **The Implication for Filters:** To handle multi-band systems, we need a “coarse” band splitter sitting *ahead* of the finer channel filters. In our architecture, this is exactly what the first-level MZI lattice filter does.

4. Space-Division Multiplexing (SDM)

- **The Mechanism:** SDM goes spatial. It adds parallel paths inside a single fiber—either by using multiple cores or multiple modes per core.
- **Pushing the Limits:** In an impressive 2023 demo, NICT hit a staggering 22.9 Pb/s using a 38-core, three-mode fiber by stacking SDM on top of multi-band WDM [3].
- **The Implication for Filters:** Crucially, each of those spatial channels still runs a full WDM system. SDM doesn’t replace WDM; it multiplies the number of wavelength filters we need on the chip.

5. Hollow-Core Fibers

- **The Mechanism:** Instead of guiding light through solid glass, these fibers guide light through an air core using a nested antiresonant cladding structure.
- **Pushing the Limits:** Recent designs have pushed attenuation down to a record-breaking 0.091 dB/km at 1550 nm, finally beating the theoretical loss limit of solid silica glass [4].
- **The Implication for Filters:** Because air guidance drastically cuts down latency and nonlinear effects, we can pump higher powers into wider WDM bands—which again, demands scalable, high-performance wavelength filters.

6. Chip-to-Chip Optical I/O

- **The Mechanism:** We’re increasingly co-packaging multiple optical wavelengths directly with processors and switches to handle massive I/O bottlenecks.
- **Where the Industry is At:** Standards like the CW-WDM MSA define grids of 8 to 32 wavelengths packed into the O-band [5]. Transceivers like TeraPHY use microring banks to handle these dense grids.
- **The Implication for Filters:** In this regime, filter footprint and the raw number of wavelengths you can squeeze onto a single fiber take priority over long-distance reach.

The Bottom Line: Whether we are scaling up modulation, spreading across new bands, adding spatial cores, or hooking up chiplets, *every single one of these dimensions* ultimately relies on splitting and combining wavelengths on a chip. As channel counts explode, relying on just one type of integrated filter isn’t cutting it anymore. That’s why the next section critically compares the four leading filter types to figure out how to combine them effectively.

1.2 The Toolbox: Four Integrated Wavelength Filters

The Core Idea: If you look under the hood of these four filters, the main difference boils down to *how the light travels* from input to output:

- **Lattices** and **AWGs** use a defined, finite set of forward paths.
- **Rings** and **CDCs** rely on feedback loops.

This fundamental structural difference dictates everything else: the shape of the passband, whether the filter repeats periodically, and how sensitive it is to manufacturing quirks.

1.2.1 Decoding “Filter Types”

Finite Impulse Response (FIR)

- **How it Works:** In an FIR filter, light takes a *finite number of paths* to reach the output, and crucially, *no light ever loops back*. The output is just a weighted sum of delayed copies of the input signal.
- **The Payoff:** Because we control the weights (via coupler ratios), we can directly sculpt the passband. Want a perfectly flat top, incredibly steep edges, or linear phase? FIR can do it.
- **The Catch:** Since we aren’t storing any energy, building a sharper filter edge requires adding more physical paths, which makes the device long and bulky.

Infinite Impulse Response (IIR)

- **How it Works:** IIR filters use *feedback*. A portion of the light continuously circulates and adds back into itself, theoretically forever.
- **The Payoff:** Feedback allows a tiny component to create a massive, sharp resonance because the trapped light interferes with itself over and over.
- **The Catch:** The passband is stuck with a peaked, Lorentzian shape (no flat tops here). Storing energy also makes the filter incredibly touchy—slight changes in temperature, loss, or manufacturing dimensions can knock it off tune.

Periodic vs. Single-Band

- **Periodic Filters:** If a filter relies on a set delay difference, its response repeats every *Free Spectral Range (FSR)*. If it selects a specific wavelength, it will also select any other wavelength that differs by exactly one phase cycle.
- **Single-Band Filters:** Filters built around gratings only respond near one specific, phase-matched wavelength. They give you exactly *one* passband, whose width is dictated by the grating’s physical properties.

With this framework in mind, let’s look at the specific components we’re working with.

1.2.2 The MZI Lattice Filter

What it is: A chain of directional couplers connected by pairs of unequal-length waveguide arms. Think of it as a series of Mach-Zehnder interferometers (MZIs) where each stage splits light into the next.

Filter Type: FIR, Periodic.

- Since light only moves forward, the response is a finite sum of delays. By tweaking the couplers, we can design flat-top “half-band” filters that perfectly split a wavelength comb into alternating channels.
- It repeats every FSR, governed by the length difference in the arms.

Pros & Cons:

- **Pros:** Excellent for flat passbands and low loss. Since it doesn’t store energy, it’s very robust against manufacturing variations.
- **Cons:** You need multiple stages to get sharp edges. Furthermore, if the coupler ratios drift with wavelength or fabrication errors, isolation suffers.

Where the Industry Uses it: It’s the go-to architecture for silicon-photonics CWDM multiplexers and textbook DWDM interleavers.

1.2.3 The Microring Resonator

What it is: A tiny, closed waveguide loop sitting next to one or two straight bus waveguides. When light at the resonant wavelength enters, it builds up inside the ring and drops out the other side.

Filter Type: IIR, Periodic.

- It feeds light back into itself, making it an all-pole (IIR) filter with a peaked Lorentzian passband.
- Resonances repeat every FSR, dictated by the ring’s circumference.

Pros & Cons:

- **Pros:** Unbeatable footprint. A micrometer-scale ring can isolate a line just tens of picometers wide. Scaling up simply means adding more rings.
- **Cons:** The periodic FSR limits how many channels you can multiplex on a single bus. Plus, the extreme sensitivity to temperature and nanometer-scale width errors means they almost always require active thermal tuning in the real world.

Where the Industry Uses it: Rings are the undisputed kings of dense optical I/O (like TeraPHY chiplets) and are widely explored for DWDM filter banks. Our proposal explores lifting the FSR limit by combining rings with grating filters.

1.2.4 The Arrayed Waveguide Grating (AWG)

What it is: An AWG consists of an input slab, an array of waveguides with linearly increasing lengths, and an output slab. The length steps tilt the phase front of the light depending on its wavelength, causing the output slab to focus different wavelengths into different output waveguides.

Filter Type: FIR, Periodic, Multi-Channel.

- Each arm acts as a delayed path (FIR tap).
- Because every output samples the same taps but with a different linear phase, the AWG is essentially performing a discrete Fourier transform in hardware—it’s an entire filter bank in one go.
- It repeats every FSR (one diffraction order).

Pros & Cons:

- **Pros:** Can handle N channels simultaneously with uniform loss and shape.
- **Cons:** The natural passband is Gaussian (flattening it costs insertion loss), crosstalk is easily ruined by random phase errors in the arms, and it gets physically massive as channel spacing shrinks.

Where the Industry Uses it: Standard DWDM multiplexers in long-haul telecom, with compact silicon versions scaling to 1×8 and 1×16 channels [6].

1.2.5 The Contra-Directional Coupler (CDC)

What it is: Two different waveguides running side-by-side with a grating corrugated between them. At a very specific wavelength, the grating kicks forward-traveling light from one guide into backward-traveling light in the other guide, dropping it out of a port on the input side.

Filter Type: Single-Band, Distributed Feedback (IIR-like).

- The coupling happens continuously along the grating, so forward and backward light interact over the whole device. This distributed feedback makes it IIR-like.
- It’s single-band because only one wavelength satisfies the phase-matching condition.
- We can achieve flat tops and low sidelobes by “apodizing” (tapering) the grating strength.

Pros & Cons:

- **Pros:** No FSR in the band of interest, highly tolerant flat passbands, and it inherently acts as an add-drop filter without needing a bulky circulator.
- **Cons:** You cannot make a physically narrow CDC with a narrow bandwidth; sharp filtering requires very long gratings (we quantify this in Section 2.2).

Where the Industry Uses it: Currently, mostly in research and foundry PDKs for flat-top CWDM demuxes or FSR-free ring selection.

1.2.6 The Filters, Side-by-Side

Table 1. Comparison of the Four Integrated Wavelength Filters

Feature	MZI Lattice	Microring	AWG	CDC
Filter Type	FIR	IIR (all-pole)	FIR, DFT filter bank	Distributed feedback, single band
Passband Shape	Flat-top (by design)	Lorentzian	Gaussian (flat costs loss)	Flat-top (with apodization)
Periodic?	Yes	Yes	Yes	No
Natural Bandwidth	nm to tens of nm	tens to hundreds of pm	0.4 nm to 20 nm	1 nm to 20 nm
Channels/Device	2 (cascade for more)	1	N	1
Physical Size	Hundreds of μm	$\approx 10 \mu\text{m}$	Hundreds of μm to mm	Hundreds of μm
Achilles' Heel	Coupler ratios	Temperature, width	Arm phase errors	Width (shifts the band)
Industry Role	CWDM (de)mux	Optical I/O, DWDM	Telecom DWDM	PDK / research filters

The Bottom Line: Looking at the table, no single filter dominates every category. That's exactly why Section 2 takes a hybrid approach: we assign each filter to the structural level where its specific strengths shine, while keeping the AWG around as a baseline benchmark for both levels.

2 Theory & Design Choices

The Mental Model for This Chapter: Everything cascades logically. The test equipment's capabilities lock in our wavelength choices. Those wavelengths dictate which filter architecture is best for each level. The filter's job then spits out a short list of required parameters, and each parameter is nailed down by one clean equation.

Here is how we build it up:

- **Section 2.1:** We pick the wavelengths.
- **Section 2.2:** We map components to those wavelengths and lay out the test structures.
- **Section 2.3:** We catalog the parameters each component needs.
- **Section 2.4:** We derive the math to calculate those parameters.

2.1 Wavelength Selection

The Core Idea: We're running four "coarse" channels spaced by 10 nm around 1550 nm. Inside the 1545 nm coarse channel, we pack four "fine" channels locked to the standard ITU 200 GHz grid. Both of these grids are strategically placed right where our grating couplers are most efficient, and the fine channels are spaced so the test laser can easily resolve them.

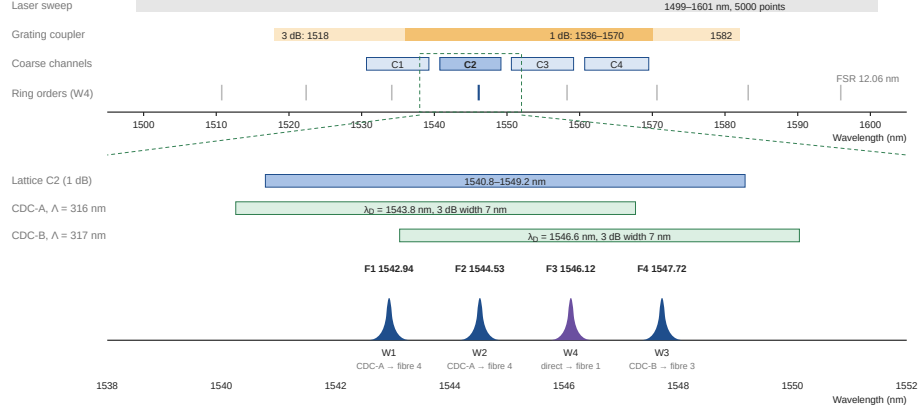


Fig. 1. Our wavelength strategy mapped against the test window. Top: the laser’s sweep range, the grating coupler’s “sweet spot” band, our coarse channels, and repeating ring orders. Bottom: zooming in on coarse channel C2 to see the CDC bands and the fine channels nestled inside.

1. The Test Window Constraints

- The openEBL test station sweeps a TE-polarized laser from 1499 to 1601 nm across 5000 discrete points [7]. This gives us a wavelength resolution (step size) of:

$$\delta\lambda_s = \frac{1601 - 1499}{5000 - 1} \text{ nm} \approx 20.4 \text{ pm}.$$

- We get light on and off the chip using the PDK’s standard grating coupler (GC_TE_1550_8degOxide_BB). The coupler has a curved transmission profile that peaks around 1550 nm and drops off at the edges [7].

2. Coarse Spacing: 10 nm (Not the Standard 20 nm)

- Why deviate from the standard 20 nm CWDM grid [2]? Every signal we measure goes through *two* grating couplers (in and out). If we spread our channels over a 60 nm span (1510–1570 nm) like CWDM, the edge channels would suffer a massive ~10 dB penalty just from the couplers.
- By squeezing our grid to 10 nm spacing (1530–1570 nm), the worst-case penalty drops to a manageable 2.6 dB.

3. Coarse Centers

- We center the grid perfectly on the coupler’s peak at 1550 nm, giving us our channels: **C1–C4 = 1535, 1545, 1555, 1565 nm.**
- In a real-world deployment, these exact 10 nm sub-bands would act as the first sorting stage for an extended-C-band DWDM grid.

4. Fine Grid: ITU 200 GHz Inside C2

- For the fine level, we adopt the ITU-T G.694.1 standard [2], placing channels at 194.3, 194.1, 193.9, and 193.7 THz.
- Using $\lambda = c/f$, these translate to **F1–F4 = 1542.94, 1544.53, 1546.12, and 1547.72 nm.**
- The spacing between them ($\Delta\lambda_{\text{grid}}$) is 1.593 nm.

- We nest these inside the C2 coarse channel (1545 nm) because it sits on the absolute flattest part of the grating coupler’s response curve. The four fine channels span just 4.78 nm, which fits beautifully inside C2’s 8.5 nm flat-top bandwidth (which we’ll prove in Section 2.4.4).

5. Fine Linewidth: Resolvable but Isolated

- **Lower Bound:** To trust our measurements, a resonance peak needs to span at least five data points from our laser sweep. That sets a minimum FWHM of ≈ 0.102 nm.
- **Upper Bound:** We can’t let neighboring channels bleed into each other, so the FWHM needs to be less than a quarter of the channel spacing (≤ 0.398 nm).
- **The Sweet Spot:** We chose an **FWHM of 0.2 nm**. This guarantees nearly 10 laser points per peak and ensures a healthy -24 dB of isolation between neighbors (derived in Section 2.4.5).

2.2 Component Mapping & Topology

The Core Idea:

- The **MZI lattice** handles the wide, flat coarse filtering.
- The **microrings** handles the ultra-narrow fine filtering.
- **CDCs** step in as “order selectors” in the test layout to kill repeating ring resonances.
- The **AWG** serves purely as our simulated baseline to judge the others against.
- The physical layout of our test structures is heavily dictated by the 1-in, 3-out fiber array we have to use.

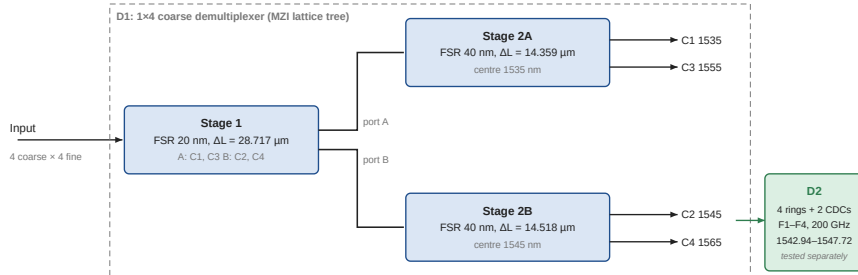


Fig. 2. The two-level architecture in action. D1 is the coarse MZI lattice tree; D2 is a bank of fine microrings.

1. The Coarse Level: MZI Lattice

- **The Job:** We need a filter that creates a flat, 10 nm wide passband with low insertion loss and high tolerance to manufacturing errors.
- **The Tool:** The MZI lattice is perfect here because its FIR nature allows us to mathematically synthesize the exact flat-top shape we need by tuning the coupler ratios (Section 1.2.1).
- **Why Not the AWG?** An AWG could do the 1×4 split natively, but its passband is naturally rounded (Gaussian). We keep it as a benchmark rather than the primary tool.

2. The Fine Level: Microrings

- **The Job:** We need a 0.2 nm linewidth filter that doesn’t consume the entire chip.
- **The Tool:** Microrings easily hit this spec with a radius of just a few micrometers.

- **Why Not a CDC?** A CDC physically cannot do this. Because of how grating physics works, narrowing the bandwidth requires lengthening the grating ($\Delta\lambda_{\min} \approx \lambda^2/(n_g L)$) [8]. To hit 0.2 nm, a CDC would need to be nearly 3 mm long—almost five times the width of our entire allocated cell!

3. Order Selection for Testing: The CDC

- **The Problem:** Rings repeat their resonances every Free Spectral Range (FSR). If we just put a ring on a bus, we’d get unwanted channels leaking through.
- **The System Solution:** In a fully integrated system, the coarse lattice filter naturally chops off these repeating orders before they ever reach the rings.
- **The Test Solution:** We can’t fit both the lattice and the rings in a single test block. So, to test the rings independently, we use a CDC as a stand-in. Because the CDC is a single-band filter with no FSR of its own, it easily passes the specific ring order we want and kills the rest.

4. The Benchmark: AWGs on Both Grids

- We designed **AWG-C** to compete head-to-head with the lattice tree, and **AWG-F** to compete with the ring bank.
- Since they process the exact same wavelength grids, the comparison is apples-to-apples. We evaluate them strictly in simulation since our physical chip real estate is maxed out by the main devices.

5. Test-Structure Topology: Working with What We Have

- The openEBL facility uses a fiber array with four fibers spaced $127\ \mu\text{m}$ apart. The laser inputs on fiber 2, leaving us one detector above and two below [7]. We have to bend our designs to fit this 1×3 constraint.
- **D1 (Lattice Tree):** A standard 1×4 lattice has four outputs, but we only have three detectors.
 - Solution: We map channels C1, C2, and C4 to the available fibers and simply terminate C3 into the slab (Fig. 3).
- **D2 (Rings + CDC):** CDCs reflect their dropped light *backwards* toward the input.
 - Solution: We route the CDC outputs back past the rings. To avoid waveguide crossings, we place ring W4 above the main bus. We map F3 (straight drop) to fiber 1, F4 (via CDC-B) to fiber 3, and multiplex F1 and F2 (via CDC-A) into fiber 4 (Fig. 4).

2.3 Parameter Selection Matrix

The Core Idea: We aren’t guessing numbers here. Each component has a handful of free parameters, and every single one is locked down by a physical requirement we set in Section 2.1 and Section 2.2. The table below acts as our roadmap.

2.4 The Math: Formulas and Derivations

2.4.1 Waveguide Fundamentals: Effective & Group Indices

The Core Idea: It’s the *group index* (n_g)—not the effective index (n_{eff})—that dictates every Free Spectral Range (FSR) on this chip. To get n_g , we must account for the dispersion of n_{eff} across wavelengths.

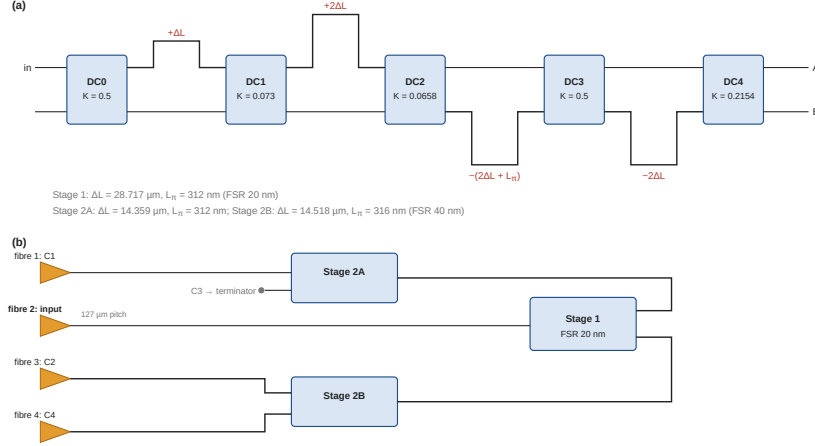


Fig. 3. (a) The core 2-channel lattice cell. (b) Device D1: The folded 1×4 lattice tree fitted to the grating-coupler array (C3 is terminated).

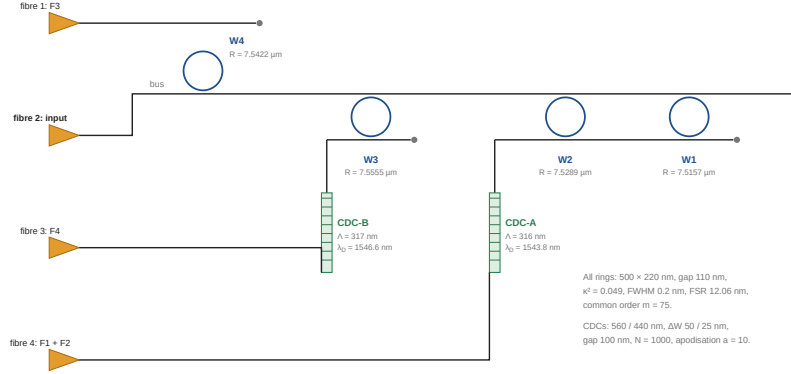


Fig. 4. Device D2: Four rings tuned by radius on a single bus. W4 drops F3 directly. W3 drops F4 via CDC-B. W1 and W2 share a drop via CDC-A.

- **Group Index:** Derived from the frequency derivative of the propagation constant [9]:

$$n_g = n_{\text{eff}} - \lambda \frac{dn_{\text{eff}}}{d\lambda}$$

- **Compact Model:** Expanding n_{eff} around $\lambda_0 = 1.55 \mu\text{m}$ yields a second-order fit that directly gives n_g and the dispersion parameter D [9]:

$$n_{\text{eff}}(\lambda) = a_0 + a_1(\lambda - \lambda_0) + a_2(\lambda - \lambda_0)^2$$

$$\Rightarrow n_g(\lambda_0) = a_0 - a_1\lambda_0$$

$$D = -\frac{2a_2\lambda_0}{c}$$

- **Working Values:** From MODE sweeps, we extract $n_{\text{eff}} \approx 2.44$ and $n_g \approx 4.18$ at 1550 nm , leading to the linear approximation used for pre-sizing:

$$n_{\text{eff}}(\lambda) \approx 2.44 - 1.1226(\lambda - 1.55 \mu\text{m})$$

Table 2. Parameter Selection Matrix

Component	Parameter	Fixed by
Strip Waveguide	$n_{\text{eff}}(\lambda), n_g(\lambda)$	Geometry (500×220 nm, TE); this feeds <i>everything</i> below.
Unbalanced MZI	ΔL	Needing enough fringes in the coupler band to fit n_g .
Directional Coupler	gap g , length L_c	The specific power splitting ratio K required at each lattice stage.
Lattice Stage	unit delay ΔL	Forcing the stage FSR to be $2^s \times$ the channel spacing.
	exact ΔL (integer m)	Centering the port-A passband directly on its target channel.
	L_π	The half-band phase offset requirement.
	coupler set $K_0..K_4$	Optimizing for a flat top and maximum isolation at the center.
Microring	radius R	Pushing the FSR wider than the 10 nm coarse spacing.
	coupling κ^2 (gap g_r)	Hitting our target FWHM of 0.2 nm.
	ΔW_i or ΔR_i	Tuning the resonances to exactly hit F1–F4.
CDC	period Λ	Centering the drop band perfectly on the ring channels.
	κ_g , length $N\Lambda$	Making the bandwidth wide enough to cover the channels at ~100% drop.
	width asymmetry $w_1 \neq w_2$	Killing unwanted self-Bragg reflections outside the main band.
Ring + Filter	channel offsets	Aligning the flat top and ensuring strong rejection of unwanted orders.
AWG	order m , ΔL	Ensuring $\text{FSR} \geq 2N\Delta\lambda$ (and > 10 nm for the fine AWG).
	focal length R_a	Mapping exactly one channel spacing to one output pitch.
	number of arms M	Capturing $\geq 99.7\%$ of the input beam's power.

- **Drift Law:** A small index change δn shifts any filter resonance by [9]:

$$\delta\lambda = \frac{\lambda \delta n}{n_g}$$

(For example, applying the thermo-optic coefficient $dn/dT = 1.8 \times 10^{-4} \text{ K}^{-1}$ [10] sets an upper bound on thermal drift of $\approx 0.067 \text{ nm/K}$ [11].)

2.4.2 Extracting the Group Index (MZI Check)

The Core Idea: We can physically measure n_g on a fabricated chip using a simple unbalanced MZI.

- By measuring the wavelength spacing (FSR) between two adjacent transmission peaks (λ_m and λ_{m+1}), we get the group index directly [9]:

$$n_g = \frac{\lambda_m \lambda_{m+1}}{\Delta L (\lambda_m - \lambda_{m+1})} \approx \frac{\lambda^2}{\text{FSR} \cdot \Delta L}$$

- We include an optional 100 μm MZI on our mask specifically to verify our simulated $n_g \approx 4.18$.

2.4.3 Sizing the Directional Couplers

The Core Idea: A directional coupler works by splitting light into an “even” and “odd” supermode. As these modes travel along the coupler, they slip out of phase, pushing power from one waveguide to the other.

- The total power crossed over (K) depends on the total phase slip, which happens in both the straight section (L_c) and the S-bends (equivalent length L_0).
- **The Design Equation:** To get a specific splitting ratio K , we set the straight length to [9]:

$$L_c = \frac{2L_x}{\pi} \arcsin(\sqrt{K}) - L_0$$

(Where L_x is the length required for a full 100% cross-over).

- We run an EME (Eigenmode Expansion) sweep over L_c to extract the exact L_x (37.18 μm) and L_0 (2.73 μm) for our specific PDK geometries.

2.4.4 Synthesizing the MZI Lattice Filter

The Core Idea: The lattice filter is an FIR filter created by cascading delays and directional couplers.

- **Setting the Spacing:** The unit delay length (ΔL) dictates the FSR of that stage. To separate channels, we set the delay to $\Delta L = \lambda^2 / (n_g \cdot 2^s \cdot \Delta\lambda_{\text{ch}})$ [12].
- **Centering the Band:** We shift the band to perfectly align with our target channel by rounding the delay to an exact integer multiple of the wavelength ($m = \text{round}(n_{\text{eff}} \Delta L / \lambda)$) [12].
- **Flattening the Top (The Minimax Optimization):** The standard “textbook” coupler ratios [12] give decent isolation (−14 dB) but degrade quickly off-center. We ran a Nelder-Mead numerical optimization to find a custom set of coupler ratios (Case C) that maintains a flat top and guarantees a worst-case leakage of **−25.9 dB across a ± 3 nm window** around the channel center.

– Our optimized coupling ratios (K) are: [0.500, 0.073, 0.066, 0.500, 0.215].

2.4.5 Microring add-drop resonator

Mechanism: Light from the bus makes repeated round trips in the ring. On resonance, returning copies arrive in phase and drain into the drop port. Off resonance, they cancel and stay on the bus.

- **Free Spectral Range (FSR):** Set by the radius R and group index n_g [13]. To exceed the 10 nm coarse spacing, we choose $R = 7.5 \mu\text{m}$.

$$\text{FSR} = \frac{\lambda^2}{n_g L} = 12.12 \text{ nm}$$

- **Linewidth (FWHM):** We target a 0.2 nm FWHM to balance resolutibility and isolation. This requires solving for the power coupling ratio κ^2 [13]:

$$\text{FWHM} = \frac{(1-x)\lambda^2}{\pi n_g L \sqrt{x}}$$

where $x = r^2 a$ combines the straight-through coupling r and propagation loss a . For 0.2 nm FWHM (assuming 3 dB/cm loss), we find $\kappa^2 = 0.049$.

- **Channel Tuning:** We place each ring on its target fine channel by adjusting the radius [13]:

$$\delta R = \frac{\delta \lambda n_g R}{\lambda n_{\text{eff}}} = 13.25 \text{ nm per 200 GHz}$$

- **Isolation Limitations:** A single ring provides -24.1 dB of isolation at the adjacent 200 GHz channel. Furthermore, its periodic FSR means unwanted orders will repeat across the spectrum, which necessitates the addition of a CDC.

2.4.6 Contra-directional coupler

Mechanism: A periodic corrugation transfers light between two modes when the momentum difference matches the grating's momentum $2\pi/\Lambda$. Using unequal guide widths prevents unwanted self-Bragg reflection from overlapping with the main drop band.

- **Phase Matching (Drop Wavelength):** The grating period Λ centers the drop band [14].

$$\lambda_D = (n_1 + n_2) \Lambda$$

For CDC-A (1543.97 nm), $\Lambda = 316$ nm [7]. The tuning slope is $d\lambda_D/d\Lambda = 2.87$ nm/nm, allowing us to accurately predict CDC-B's required period as 317 nm.

- **Bandwidth:** The grating strength κ_g and length L set the bandwidth [8].

$$\Delta\lambda = \frac{\lambda_D^2}{\pi \tilde{n}_g} \sqrt{|\kappa_g|^2 + \left(\frac{\pi}{L}\right)^2}$$

To hit a ~ 4 nm bandwidth at $N = 1000$, we extract a required coupling of $|\kappa_g| = 0.0201 \mu\text{m}^{-1}$.

2.4.7 Two-level selection rule

A band filter (the CDC or the coarse MZI lattice) serving a ring must hold the ring's channel inside its flat top *and* reject the ring's neighboring repeating orders [15].

$$\begin{aligned} |\lambda_F - \lambda_{\text{band}}| + \delta\lambda_{\text{diff}} &\leq B_{\text{1dB}}, \\ \text{FSR}_{\text{ring}} - |\lambda_F - \lambda_{\text{band}}| &\geq B_{\text{rej}} \end{aligned}$$

Both our CDC and Lattice configurations easily satisfy this, providing >1 nm of alignment margin.

2.4.8 Arrayed waveguide grating

Mechanism: The input slab spreads light across an array of arms, each ΔL longer than the last. This linear phase step tilts the wavefront, sweeping the focus across the output guides based on wavelength.

- **FSR and Order:** The path difference ΔL and order m set the FSR [16].

$$m = \frac{n_{\text{eff}} \Delta L}{\lambda_c} \quad \text{FSR} = \frac{\lambda_c^2}{n_g \Delta L}$$

- **Focal Length (R_a):** Converts the angular dispersion into exactly one channel spacing per output pitch d_o [16].

$$R_a = \frac{d_o n_s d \lambda_c}{n_g \Delta L \Delta \lambda_{\text{ch}}}$$

- **Number of Arms (M):** Must be wide enough to capture the diffracted input beam [17].

$$M \geq \frac{3 \theta_0 R_a}{d}$$

Applying these formulas gives us AWG-C ($m = 11$, $R_a = 60.1 \mu\text{m}$) and AWG-F ($m = 71$, $R_a = 58.5 \mu\text{m}$) as solid benchmarks.

2.4.9 Test-set constraints

A 4-coupler device fills the full height of the openEBL layout cell [7].

$$N_{\text{max}} = 1 + \left\lfloor \frac{410 - 27.2}{127} \right\rfloor = 4$$

Because the test structures require full height, Devices D1 and D2 must be placed side-by-side, which exhausts the width of the cell. This physical limitation is why the AWG benchmarks must remain in simulation.

3 Modelling and Simulation

The Core Idea: Each component is simulated on its own first, from the waveguide up, and the results then feed the two device-level INTERCONNECT simulations, D1 and D2. Where the SiEPIC PDK already has data for a part, our result is checked against it.

3.1 Strip Waveguide

The Core Idea: Every filter on the chip depends on the effective and group index of one $500 \times 220 \text{ nm}$ strip, so this is solved first (MODE FDE, TE0, SiO₂ cladding, $2 \mu\text{m}$ BOX [7]).

Index and dispersion (1500–1600 nm, 11 wavelengths; Fig. 5):

- **Compact model, λ in μm :**

$$n_{\text{eff}}(\lambda) = 2.44198 - 1.12195(\lambda - 1.55) - 0.03475(\lambda - 1.55)^2$$

- **Group index:** $n_g = a_0 - a_1 \lambda_0 = 2.44198 + 1.12195 \times 1.55 = 4.181$, confirming the 4.18 used for sizing.
- **Dispersion:** $D = -2a_2 \lambda_0 / c = +359 \text{ ps}/(\text{nm} \cdot \text{km})$.
- **Mesh:** 20 nm base with 10 nm on the core. A uniform 10 nm mesh changes n_{eff} by only 1.4×10^{-4} at four times the cost.

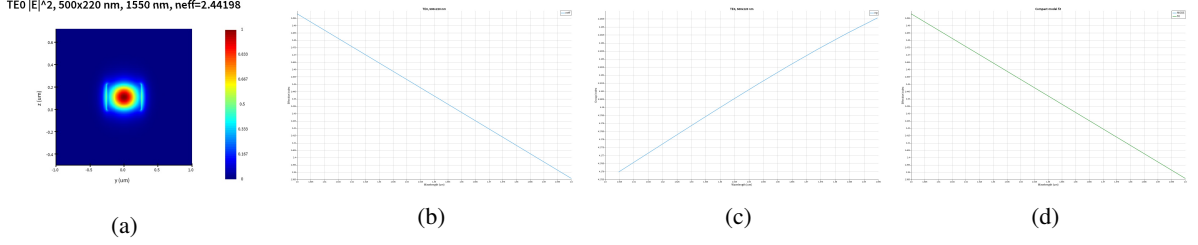


Fig. 5. 500×220 nm strip waveguide (MODE FDE). (a) TE0 $|E|^2$ at 1550 nm. (b) Effective index versus wavelength. (c) Group index versus wavelength. (d) Quadratic compact-model fit against the MODE data.

Fabrication sensitivity (at 1545 nm; Fig. 6). A resonance moves by $\delta\lambda = \lambda \delta n/n_g$ for an index change δn :

- **Width** (440–560 nm): $\partial n_{\text{eff}}/\partial W = 1.574 \times 10^{-3}$ per nm, i.e. 0.581 nm of shift per nm of width.
- **Thickness** (210–230 nm): $\partial n_{\text{eff}}/\partial H = 3.705 \times 10^{-3}$ per nm, i.e. 1.37 nm of shift per nm of thickness, 2.4 times the width sensitivity. Thickness varies slowly across the wafer, so it shifts all filters together.
- **Second mode:** at 500 nm the TE1 mode sits at $n_{\text{eff}} = 1.496$, only 0.05 above the cladding (1.444), and is cut off below 440 nm. The ring bends and the TE0 grating couplers strip it, so the guide is single-mode in practice. The CDC’s 560 nm guide supports a better-confined TE1 (n_{eff} 1.60), as in the PDK design.

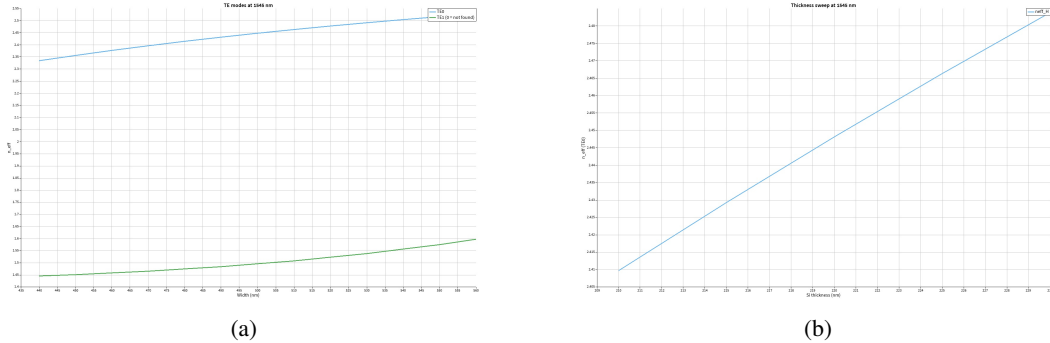


Fig. 6. (a) TE0 and TE1 effective index versus waveguide width at 1545 nm. (b) TE0 effective index versus Si thickness at 1545 nm.

Check against the PDK (waveguide data tables of the EBeam INTERCONNECT library [7]):

Table 3. Strip Waveguide: Simulation Versus PDK Data

Quantity	PDK	Ours	Difference
n_{eff}	2.44553	2.44198	−0.15 %
n_g	4.19088	4.181	−0.24 %
$\partial n_{\text{eff}}/\partial W$	$1.581 \times 10^{-3} \text{ nm}^{-1}$	$1.574 \times 10^{-3} \text{ nm}^{-1}$	−0.47 %
$\partial n_{\text{eff}}/\partial H$	$3.712 \times 10^{-3} \text{ nm}^{-1}$	$3.705 \times 10^{-3} \text{ nm}^{-1}$	−0.19 %

The 0.24 % difference in n_g moves a 12.12 nm ring FSR by only 0.03 nm, a sixth of the 0.2 nm linewidth, so our compact model is used as it is.

3.2 Bent Waveguide (Rings)

The Core Idea: The rings are tight ($R = 7.5 \mu\text{m}$), so the bend changes the index slightly and may add loss. Both are checked with a bent-mode FDE solve (Fig. 7).

- **Index:** the bent mode has $n_{\text{eff}} = 2.4426$, 0.00063 above the straight guide, and $n_g = 4.1817$. The ring FSR is then $\lambda^2 / (n_g 2\pi R) = 12.11 \text{ nm}$, against 12.12 nm from the straight-guide estimate.
- **Radiation loss:** about $2 \times 10^{-5} \text{ dB}$ per round trip, at the solver's floor. It is negligible for $R \geq 5 \mu\text{m}$.
- **Junction loss:** 0.0027 dB per straight-to-bend junction at $7.5 \mu\text{m}$ (0.0060 dB at $5 \mu\text{m}$, 0.0015 dB at $10 \mu\text{m}$).
- **So:** ring loss is set by sidewall scattering, not by bending, and the 3 dB/cm assumption stands until measured.

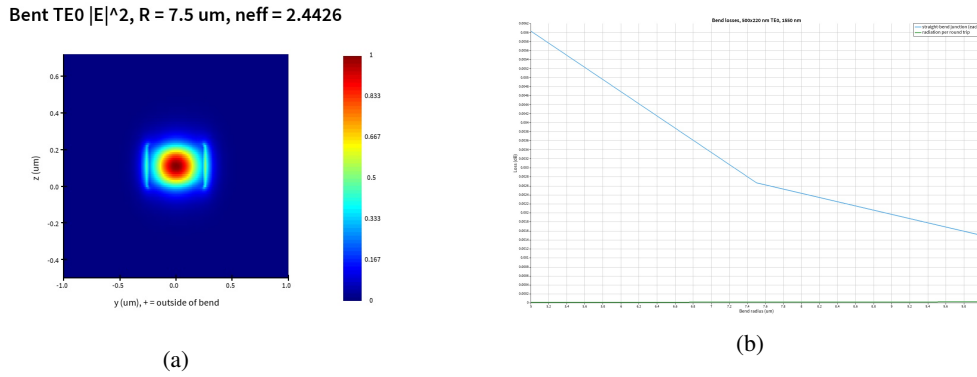


Fig. 7. (a) Bent TE0 $|E|^2$ at $R = 7.5 \mu\text{m}$. (b) Straight-to-bend junction loss and radiation loss versus radius.

3.3 Directional Couplers (Lattice Taps)

The Core Idea: All lattice couplers use the same 200 nm gap [9]; only the straight length L_c changes. FDE gives the cross-over length L_x , EME of the full coupler adds the S-bend term L_0 , and then $L_c = (2L_x/\pi) \arcsin \sqrt{K} - L_0$ for each target K .

Supermodes (FDE; Figs. 8–9):

- **Gap sweep:** $L_x = \lambda / [2(n_e - n_o)] = 23.90 / 29.94 / 37.37 / 47.07 / 58.59 \mu\text{m}$ at gaps of 150 / 175 / 200 / 225 / 250 nm (1550 nm).
- **Design point** (200 nm): $n_e = 2.45352$, $n_o = 2.43278$, so $L_x = 37.37 \mu\text{m}$. The even mode bridges the gap; the odd mode flips sign across it, with a null in the gap.
- **Wavelength:** L_x shortens with λ , so a 50/50 coupler drifts from $K = 0.40$ to 0.61 across the band.

Full coupler (EME; Fig. 10). 10 μm cosine S-bends, 10 nm mesh on both cores, L_c swept 0.05–40 μm . Fitting $K = \sin^2[\pi(L_c + L_0)/(2L_x)]$ gives:

- **Agreement:** EME and FDE L_x agree to 0.5 %. The default EME mesh gave 9 % short, so the fine core mesh is necessary.
- **Layout lengths** (optimised set, 1550 nm):

Check against the PDK coupler (ebeam_dc_te1550, same 500 nm guides and 200 nm gap [7]). The same fit to the PDK S-parameters gives $L_x = 43.01 / 39.10 / 36.01 \mu\text{m}$ at 1530 / 1550 / 1570 nm.

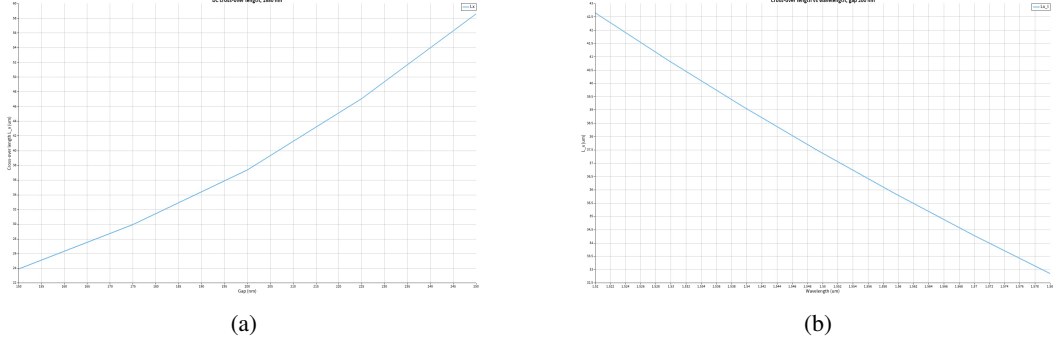


Fig. 8. Directional-coupler cross-over length (a) versus gap at 1550 nm and (b) versus wavelength at a 200 nm gap.

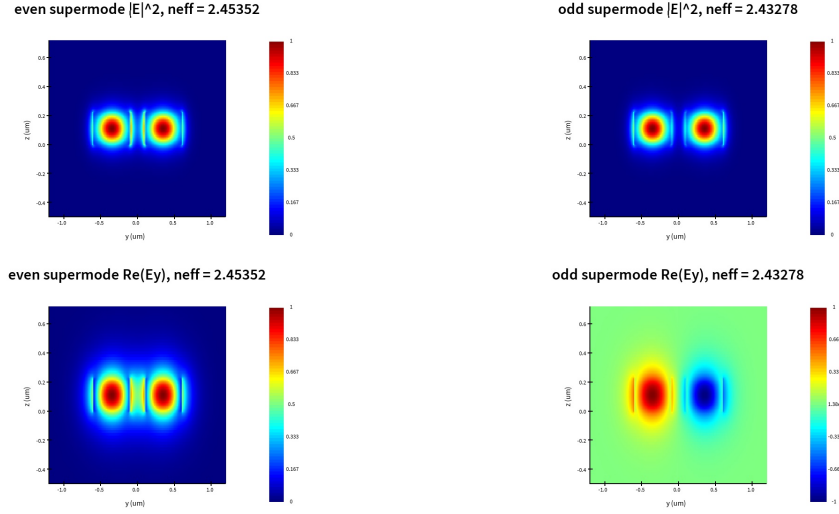


Fig. 9. Even (left) and odd (right) TE supermodes of the lattice directional coupler (two 500×220 nm guides, 200 nm gap, 1550 nm). Top: $|E|^2$. Bottom: $\text{Re}(E_y)$.

Table 4. Directional-Coupler EME Fit

λ (nm)	L_x (μm)	L_0 (μm)	bar + cross
1530	40.59	2.70	0.986–0.999
1550	37.18	2.73	0.987–0.999
1570	34.11	2.77	0.988–0.999

Table 5. Layout Coupling Lengths (Optimised Set, 1550 nm)

K	0.500	0.0730	0.0658	0.500	0.2154
L_c (μm)	15.86	3.74	3.41	15.86	8.69

- **Our L_x is 4.4 % shorter.** If the PDK is right, a 50/50 tap built on our value lands at $K = 0.535$.
- **The dispersion agrees:** L_x falls $0.175 \mu\text{m}/\text{nm}$ in the PDK and $0.163 \mu\text{m}/\text{nm}$ in ours.

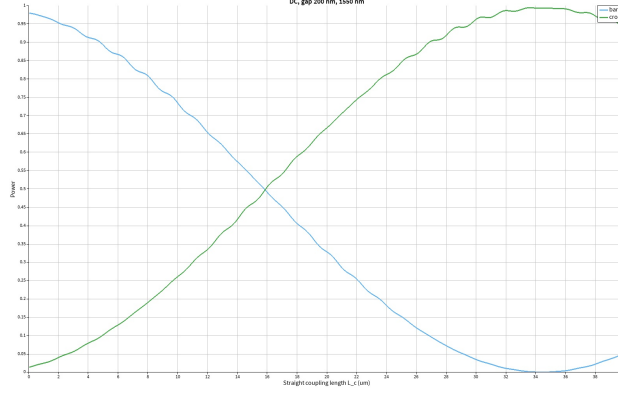


Fig. 10. EME bar and cross power versus straight coupling length, gap 200 nm, 1550 nm. At $L_c = 0$ the S-bends alone cross over 1.4 %.

3.4 Lattice Tree (D1)

The Core Idea: The three stages of the 1×4 tree are simulated together in INTERCONNECT, with the waveguide compact model in the delays and each coupler modelled twice: as an ideal, wavelength-flat coupler (tests the filter design) and as the 4-port EME S-parameters (adds the real $K(\lambda)$, coupler phase and 1–2 % loss).

Table 6. Lattice Stage Delays

Stage	FSR	ΔL	L_π	Ports
1	20 nm	28.717 μm	0.312 μm	A: C1, C3 · B: C2, C4
2A	40 nm	14.359 μm	0.312 μm	A: C1 · B: C3
2B	40 nm	14.518 μm	0.316 μm	A: C2 · B: C4

Three coupler sets are compared: **A**, the published set $K = [0.50, 0.13, 0.12, 0.50, 0.25]$ [12]; **B**, the same with 0.437 in place of 0.25; and **C**, our optimised set $K = [0.500, 0.073, 0.066, 0.500, 0.215]$.

Table 7. Simulated 1×4 Lattice-Tree Performance

Set	Couplers	Peak loss (dB)	Worst leakage within ± 3 nm, C1 / C2 / C3 / C4 (dB)
A	ideal	0.09–0.12	–14.3 / –14.3 / –14.3 / –14.3
A	EME	0.31–0.55	–15.3 / –14.8 / –15.6 / –15.5
B	ideal	0.09–0.10	–13.2 / –12.7 / –12.1 / –11.6
B	EME	0.29–0.62	–13.2 / –13.3 / –12.6 / –9.6
C	ideal	0.09–0.11	– 25.2 / – 24.1 / – 22.3 / – 20.9
C	EME	0.40–0.47	– 25.3 / – 22.5 / – 19.4 / – 15.2

- **Centring:** every passband lands within 0.1 nm of its channel, with 1 dB widths of 8.3–8.9 nm.
- **Set C is the best design**, beating A by 10.0, 7.7 and 3.8 dB at C1–C3 with real couplers.
- **Its advantage fades toward C4.** With ideal couplers it degrades from –25.2 to –20.9 dB, probably from waveguide dispersion, since each stage is centred on one reference channel. With real couplers C4 falls to –15.2 dB, level with set A, because the 50/50 coupler drifts from $K = 0.501$ at 1550 nm to 0.665 at 1600 nm. Closing that gap needs couplers with flatter $K(\lambda)$, not a different K set.

- **Loss:** 0.3–0.6 dB through ten couplers.
- **Outputs:** C1 → fibre 1, C2 → fibre 3, C4 → fibre 4; C3 terminated.

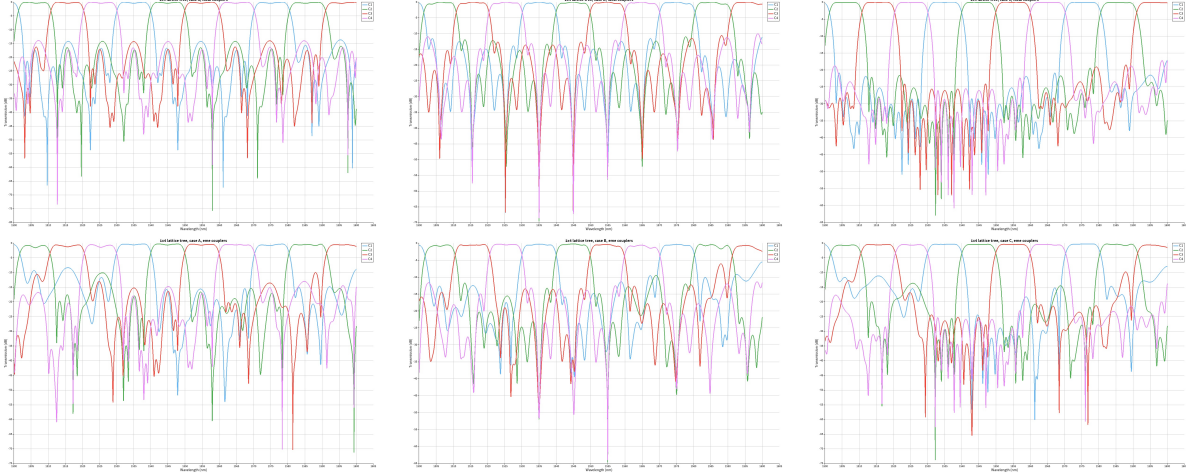


Fig. 11. INTERCONNECT transmission of the 1×4 lattice tree (ports C1–C4), 1500–1600 nm. Columns: sets A, B, C. Top row: ideal couplers. Bottom row: EME couplers.

3.5 Unbalanced MZI (Optional n_g Check)

A 100 μm path difference gives an FSR of $\lambda^2/(n_g \Delta L) = 5.75$ nm, about ten fringes in 60 nm. It is simulated in INTERCONNECT with PDK Y-branches; the measured FSR returns n_g for comparison with 4.181.

3.6 Ring Resonators

The Core Idea: The four rings differ only in radius, from 7.5157 to 7.5555 μm in 13.3 nm steps; each step moves the resonance by one 1.593 nm channel. Everything else is common: a 110 nm gap, $\kappa^2 = 0.049$ per coupler, and a 0.2 nm linewidth.

Table 8. Ring Radii and Routing in Device D2

Ring	Channel	Target λ	Radius	Drop routed to
W4	F3	1546.12 nm	7.5422 μm	fibre 1, direct
W3	F4	1547.72 nm	7.5555 μm	CDC-B → fibre 3
W2	F2	1544.53 nm	7.5289 μm	CDC-A → fibre 4
W1	F1	1542.94 nm	7.5157 μm	CDC-A → fibre 4

Radii. All four rings run at order $m = 75$, with circumference $L = m\lambda_F/n_{\text{eff,bent}}$, where $n_{\text{eff,bent}}$ is the compact-model index plus the 0.00063 bend correction. For W2: $L = 75 \times 1.544526/(2.44812 + 0.00063) = 47.3055 \mu\text{m}$, so $R = L/2\pi = 7.5289 \mu\text{m}$.

- **The bend correction matters:** sizing with the straight-guide index put every resonance 0.22 nm red of its channel. With it, a bent-ring FDE solve at each radius puts all four within 0.01 nm ($n_g = 4.182$, FSR = 12.06 nm).

- **Radius rather than width:** one channel needs 13.3 nm of radius or 2.74 nm of width. On the 1 nm layout grid, width tuning quantises the channel to 36 % of a spacing, radius tuning to 7.5 %. All rings are therefore 500 nm wide.

Table 9. Parameters Common to All Four Rings

Common parameter	Value
Gap, bus–ring and ring–drop	110 nm
Power coupling per coupler, κ^2	0.049
Linewidth (FWHM), Q_L , finesse	0.2 nm, 7725, 60.6
FSR	12.06 nm
Drop loss	−0.28 dB
Tuning	0.120 nm per nm of radius; 0.591 nm per nm of width
Adjacent channel	1.593 nm = $8.0 \times \text{FWHM} \rightarrow -24.1$ dB for a single ring

Setting the gap. Two independent routes both give 110 nm:

1. **FDE model:** supermodes of two straight guides give $\Delta n(g) = 0.1260 e^{-g/110.9 \text{ nm}}$. Integrating along the curved gap, $\kappa = \sin \int \pi \Delta n(g(x)) / \lambda dx$, gives $\kappa^2 = 0.049$ at 106.6 nm (Fig. 12a). Checked against the PDK’s 3D-FDTD half-ring data [7], the model runs low, more so at small radius:

Table 10. Ring Coupling: FDE Model Versus PDK Half-Ring Data (1550 nm)

R	Gap	PDK κ^2	FDE model κ^2	Model / PDK
10 μm	150 nm	0.0300	0.0300	1.00
10 μm	100 nm	0.0763	0.0729	0.96
5 μm	100 nm	0.0415	0.0366	0.88
3 μm	100 nm	0.0267	0.0218	0.82

Correcting by the interpolated ratio at $R = 7.5 \mu\text{m}$ (about 0.92) raises the gap to 110 nm.

2. **PDK scaling:** the PDK points follow $\kappa^2 \propto R^{0.88}$ and $\kappa^2 \propto e^{-g/53.6 \text{ nm}}$. At $R = 7.5 \mu\text{m}$ this gives $\kappa^2 = 0.0594$ at 100 nm, so $g = 100 + 53.6 \ln(0.0594/0.049) = 110$ nm.

varFDTD was rejected for this job. At the PDK calibration point ($R = 5 \mu\text{m}$, gap 100 nm) it gives $\kappa^2 \approx 0.085\text{--}0.105$, 2.3 times the PDK value, with a ± 12 % ripple and through + cross above 1 (Fig. 12b). The 2.5D method overstates coupling across a curved gap; a 3D FDTD half-ring would be the reference.

3.7 Contra-Directional Couplers

The Core Idea: The CDCs are the PDK cell (560/440 nm guides, 50/25 nm corrugation, 100 nm gap, $N = 1000$ [7]) at two periods: $\Lambda = 316$ nm (CDC-A, passes F1 and F2) and 317 nm (CDC-B, passes F4). Two things are checked: where the drop band lands, and how strong the grating is in the cell we actually lay out.

Drop wavelength (FDE supermodes of the unperturbed pair; Fig. 13). The drop occurs where $\lambda = (n_1 + n_2)\Lambda$:

- **Agreement:** within 0.25 nm of the PDK, a sixth of a channel spacing. The tuning slope is 2.88 nm per nm of period (PDK 2.87), so the CDC-B prediction holds.

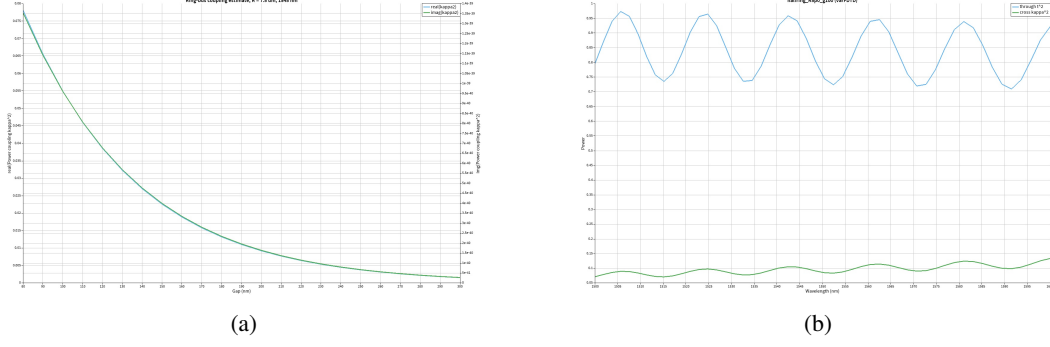


Fig. 12. (a) Ring–bus power coupling versus gap from FDE supermodes and the curved-gap integral, $R = 7.5 \mu\text{m}$, 1545 nm. (b) varFDTD half-ring at the PDK calibration point ($R = 5 \mu\text{m}$, gap 100 nm); note the ripple and a cross power about $2.3\times$ the PDK 3D-FDTD value (0.0415).

Table 11. CDC Drop and Self-Bragg Wavelengths

Period	λ_D (ours)	λ_D (PDK)	Self-Bragg reflections
316 nm (CDC-A)	1543.72 nm	1543.97 nm	1580.96 / 1509.85 nm
317 nm (CDC-B)	1546.60 nm	—	1584.07 / 1512.54 nm

- **Self-Bragg reflections** sit 34–37 nm from the drop band, well outside the fine band but inside the laser sweep, so they will appear in the measured spectra.

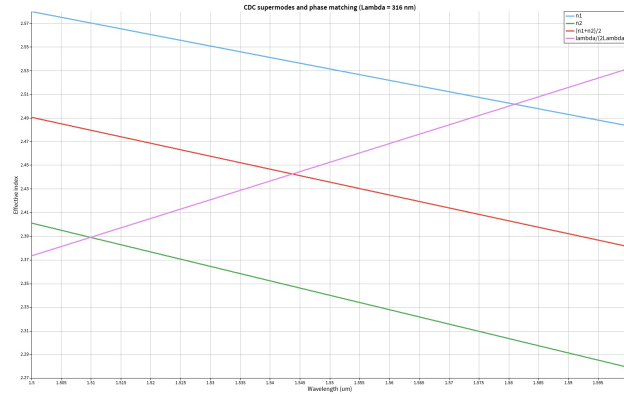


Fig. 13. CDC supermode indices n_1, n_2 and the phase-matching line $\lambda/(2\Lambda)$ for $\Lambda = 316 \text{ nm}$. The drop occurs where $(n_1 + n_2)/2$ crosses it (1543.7 nm); each self-Bragg reflection where one n_i crosses it.

Grating strength (coupled-mode transfer matrix [8], 100 segments; Fig. 14).

1. **Calibrate to the PDK model:** its 4.09 nm bandwidth is reproduced with $\kappa_0 = 0.0227 \mu\text{m}^{-1}$.
2. **Apply it to the layout cell.** The PDK model and the PCell use different apodisation formulas, $e^{-a(i-N/2)^2/N^2}$ and $e^{-\frac{1}{2}[2a(i-N/2)/N]^2}$, so the same index a gives a much narrower Gaussian in the PCell:

So the PDK S-parameters describe an $a \approx 3$ device, not the $a = 10$ cell in the layout.

3. **Cross-check with EME of one grating period:** the reflection per period is $|r_{12}| = 0.00947$, so $\kappa_0 = \text{atanh}(0.00947)/0.316 \mu\text{m} = 0.0300 \mu\text{m}^{-1}$. This is 32 % higher than the calibrated value, as expected: EME sees sharp drawn teeth, while fabricated teeth are rounded. With this upper value

Table 12. CDC Drop Versus PCell Apodisation Index ($\kappa_0 = 0.0227 \mu\text{m}^{-1}$)

PCell apodisation a	Peak drop	3 dB width
10 (default; our layout)	−2.90 dB	6.15 nm
5	−0.48 dB	4.65 nm
3	−0.04 dB	4.25 nm
2	−0.003 dB	4.05 nm

the $a = 10$ cells drop −1.62 dB (CDC-A, 1543.75 nm) and −1.61 dB (CDC-B, 1546.62 nm), each 7.0 nm wide.

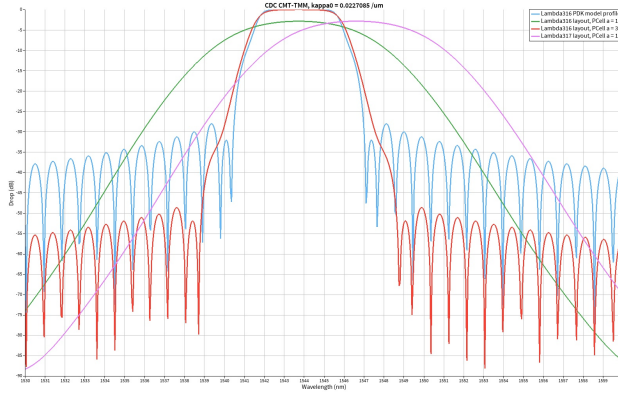


Fig. 14. CMT drop spectra, $\kappa_0 = 0.0227 \mu\text{m}^{-1}$, $N = 1000$. Blue: PDK model profile. Green: PCell $a = 10$. Red: PCell $a = 3$. Magenta: $\Lambda = 317$ nm with PCell $a = 10$.

Choice: keep $a = 10$, because it is easier to fabricate. The teeth are 50 nm and 25 nm deep at the grating centre; the apodisation sets how many taper into the 2–10 nm depth range that e-beam renders unreliably:

Table 13. Tooth-Depth Distribution Versus Apodisation Index

a	Teeth ≥ 10 nm	2–10 nm (uncertain)	< 2 nm (plain guide)
10	136–180	74–88	746–776
3	452–598	248–298	154–250

- **Gain:** three times fewer marginal teeth, and it is the PDK default cell.
- **Cost:** a weaker (1.6–2.9 dB) and wider (6–7 nm) drop.
- **Why that is acceptable:** the CDCs sit after the rings, which already select one channel each. The CDC’s job is to clean up the rings’ drop tails, so a wider band costs isolation margin, not channel selection.

3.8 Ring Bank (D2)

The Core Idea: The four rings and two CDCs are simulated together in INTERCONNECT. The rings are coupler pairs ($\kappa^2 = 0.049$) with bent-index arcs. The CDCs are run twice: with the PDK compact model ($a \approx 3$), and with the $a = 10$ layout cell from the coupled-mode model at $\kappa_0 = 0.0300 \mu\text{m}^{-1}$ (Fig. 15).

Table 14. Device D2 Channel Response

Channel	Output	Resonance	3 dB width	Peak, PDK CDC	Peak, $a = 10$ CDC
F1	fibre 4, via CDC-A	1542.93 nm	0.182 nm	−0.42 dB	−2.09 dB
F2	fibre 4, via CDC-A	1544.53 nm	0.203 nm	−0.36 dB	−2.06 dB
F3	fibre 1, W4 drop	1546.11 nm	0.183 nm	−0.31 dB	−0.31 dB
F4	fibre 3, via CDC-B	1547.72 nm	0.163 nm	−0.36 dB	−2.12 dB

Power (dB) on each output at each channel centre, PDK CDC model ($a = 10$ in brackets where it differs):

Table 15. Device D2 Output Power at Each Channel Centre (dB)

Output	F1	F2	F3	F4
fibre 1 (W4 drop)	−29.3	−24.1	− 0.3	−24.1
fibre 3 (CDC-B)	−91.2 (−36.4)	−33.3 (−31.8)	−44.8 (−46.5)	− 0.4
fibre 4 (CDC-A)	− 0.4	− 0.4	−51.8 (−51.5)	−101.4 (−66.4)

- **Every channel lands on target** with the designed 0.2 nm linewidth.
- **The $a = 10$ CDCs cost about 1.7 dB** per CDC-routed channel, but still isolate by at least 31.8 dB.
- **The weakest output is the plain W4 drop**, with −24.1 dB at the adjacent channels.

Why the CDC stage is needed: the W4 drop. With only three output fibres, W4’s drop goes straight to fibre 1 without a CDC. It shows both limits of a single ring:

Table 16. W4 Plain-Drop Response Versus Single-Ring Prediction

W4 drop, fibre 1	Simulated	Single-ring prediction
Repeating orders, 1500–1600 nm	8 peaks at −0.3 dB, 12.06 nm apart	8–9 at −0.28 dB
Leak at adjacent channels F2 / F4	−24.1 / −24.1 dB	−24.1 dB
Leak at F1 (3.19 nm away)	−29.3 dB	−30.0 dB
Leak at coarse channels C1 / C2 / C3 / C4	−19.0 / −21.3 / −29.5 / −31.9 dB	C1: −19.5 dB

By contrast, at the other orders of W1–W3 the CDC outputs stay between −55 and −110 dB.

- **Limit 1, repeating orders:** fibre 1 shows eight equal peaks, and coarse-channel light would appear at −19 dB. In the full system the lattice’s C2 port removes C1, C3 and C4 first, so this is a test-structure problem only.
- **Limit 2, adjacent-channel leak (−24 dB):** this applies in the system too.
- **Decision: W4 stays a plain ring drop.** Fibre 1 against fibres 3 and 4 then measures both limits directly. The alternatives were a third CDC at $\Lambda = 316.75$ nm (removes Limit 1 but not Limit 2, and adds a $317 \mu\text{m}$ cell) or a narrower W4 (FWHM 0.1 nm, gap ≈ 150 nm: −30.1 dB adjacent for 0.28 dB more drop loss).

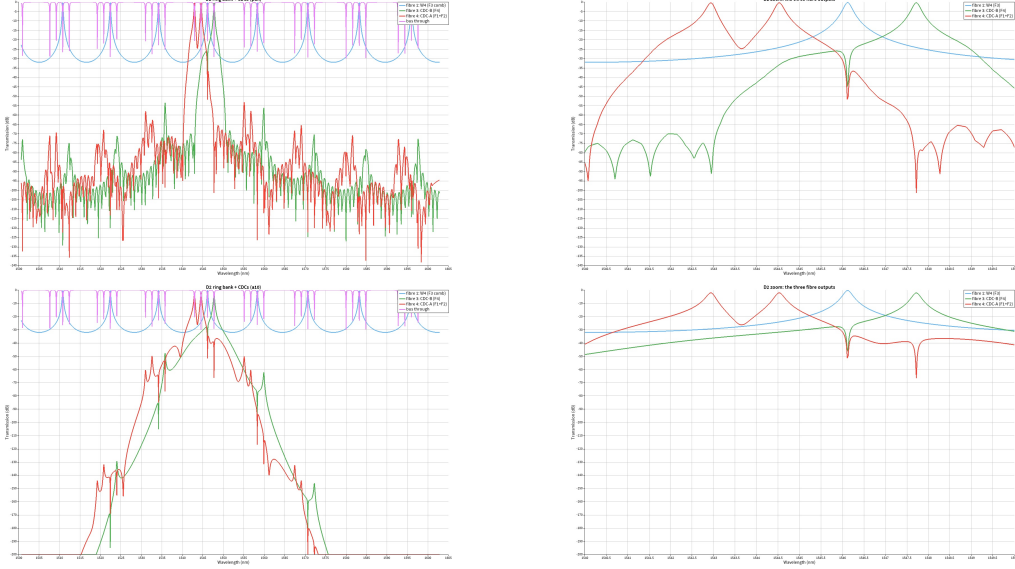


Fig. 15. INTERCONNECT response of device D2: W4 drop (fibre 1), CDC-B (fibre 3), CDC-A (fibre 4) and the bus through port. Left: full band. Right: zoom, 1540–1550 nm. Top row: PDK CDC model. Bottom row: the $a = 10$ layout CDCs.

3.9 Arrayed Waveguide Gratings (Simulation Only)

The Core Idea: Because the AWGs are too large to fit in the test cell, they are strictly used as simulated benchmarks. Their response is built by chaining the slab star-coupler (varFDTD) and the dispersive waveguide arms (scripted phase delay) into a full S-matrix in INTERCONNECT [18].

- **Coarse Level (AWG-C):** Matches the 10 nm C1–C4 grid using order $m = 11$, with an arm length difference of $\Delta L = 6.988 \mu\text{m}$. This sets an FSR of 82.25 nm and requires a focal length $R_a = 60.1 \mu\text{m}$.
- **Fine Level (AWG-F):** Matches the 1.593 nm F1–F4 grid using order $m = 71$, with $\Delta L = 44.870 \mu\text{m}$. This tightens the FSR to 12.73 nm and uses a focal length $R_a = 58.5 \mu\text{m}$.
- **Arm Count:** Both require 26–27 arms to capture the diffracted input beam.
- **Cell Constraint:** The fine-level AWG arms spread over 1122 μm , explicitly violating the 410 μm test cell constraint.

3.10 PDK Parts (No Simulation)

The Core Idea: We do not re-simulate standard passive components that have trusted, pre-calibrated S-parameter models provided directly by the SiEPIC EBeam library [7].

- **Grating Coupler (GC_TE_1550_8degOxide_BB):** $40.0 \times 27.2 \mu\text{m}$ footprint. Used on all inputs and outputs; dictates the baseline insertion loss penalty and the parabolic transmission envelope.
- **Y-Branch (ebeam_y_1550):** Used in the unbalanced MZI check.
- **Terminator (ebeam_terminator_te1550):** Absorbs the unused C3 output port of the lattice tree.

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